

1. Predict the geometry and number of unpaired spins in the following four-coordinate complexes: $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$. Explain your reasoning.
2. The octahedral structure is not the only possible six-coordinate structure. Other possibilities include a planar hexagonal structure and a triangular prism structure. In the latter, the ligands are arranged in two parallel triangles, one lying above the metal atom and the other below the metal atom with its corners directly in line with the corners of the first triangle. Show that the existence of two and only two isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is evidence against both of these possible structures.
3. Is the coordination compound $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ diamagnetic or paramagnetic?
4. In what directions do you expect the bond length and vibrational frequency of a free CO molecule to change when it becomes a CO ligand in a $\text{Ni}(\text{CO})_4$ molecule? Explain your reasoning.
5. Molecular nitrogen (N_2) can act as a ligand in certain coordination complexes. Predict the structure of $[\text{V}(\text{N}_2)_6]$, which is isolated by condensing V with N_2 at 25 K. Is this compound diamagnetic or paramagnetic? What is the formula of the carbonyl compound of vanadium that has the same number of electrons?